

What would unblock the confinement-scale region, and what was checked and fails

The map of the position note leaves one region untouched by every tool with explicit constants: the confinement scale, where the effective coupling is of order one and the effective theory must be shown to be mixing uniformly in the volume. This note records the outcome of a review of every rigorous technique known to the author that might enter that region by written derivation, so that later sessions do not re-derive the same dead ends. The conclusion is that exactly two things would unblock it, and neither is available here:

1. **A correlation inequality for $SU(3)$ lattice gauge theory** of Griffiths–Ginibre type, giving monotonicity in the coupling of the quantities that control mixing. Researched further in the research note: it would transfer control from weaker to stronger coupling only, replacing the strong-side steps and leaving the weak side untouched, so it is a simplification rather than an unblocking. This is what makes the Ising model’s intermediate regime accessible and what the abelian gauge theories have; for $SU(N)$ none of this type is known, Section 1 says where the standard constructions break, and the centre-based comparison inequalities that do exist concern the string tension rather than the vacuum-sector gap.
2. **A computer-assisted verification of the Dobrushin–Shlosman block criterion** for the Wilson interaction, or for the effective interaction at the confinement scale, as specified in the Dobrushin note §4. This is numerics of a size beyond present practice and outside this repository’s rules.

Everything else checked reduces to one of these or fails for a stated reason. References at metadata level; nothing promoted.

1. Correlation inequalities: where they break for $SU(N)$

Ising and abelian gauge theories. Griffiths’ inequalities in Ginibre’s general form (Commun. Math. Phys. 16 (1970) 310) need a cone of functions on the single-site space, closed under products, such that the duplicate-variables integral $\iint \prod_i (f_i(x) - f_i(y)) d\nu(x) d\nu(y) \geq 0$ holds. For ± 1 spins and for $U(1)$ with the cosines this holds, so the Wilson loops of $U(1)$ lattice gauge theory are monotone in β_W and the phase structure can be squeezed from both sides; the sharpness theorem of Aizenman, Barsky and Fernández (J. Stat. Phys. 47 (1987) 343) is the deepest form of this control for Ising-type models. The Coulomb phase of $U(1)$ (Guth; Fröhlich–Spencer) uses the dual representation, whose weights are positive.

Non-abelian. Three obstructions, each sufficient on its own.

- The dual representation has signs. Integrating the links of a character expansion produces recoupling coefficients, which are not nonnegative for $SU(2)$ and $SU(3)$, so the surface gas is not a positive measure and no monotonicity in β_W follows from it. The plaquette coefficients themselves are nonnegative (Wilson note §1), which is why the strong-coupling expansion is a positive gas; the signs appear at the vertices where more than two surfaces meet.
- The $O(N)$ analogy fails already at $N = 4$. $SU(2) \simeq S^3$, and $\frac{1}{2} \text{tr}(UV^\dagger)$ is the Euclidean inner product of unit quaternions, so the plaquette term is a degree-four polynomial in four $O(4)$ spins; Ginibre’s construction covers $O(2)$ and specific $O(N)$ interactions but not this one, and $SU(3)$ has no such real form at all.
- The interaction is not ferromagnetic in any sign sense: for a complex defining representation $\text{Re tr } U_p$ is not a product of functions on the links with definite sign properties.

What does exist for non-abelian groups, and why it does not reach. Comparison inequalities built on the centre are known and rigorous. Mack and Petkova (Ann. Phys. 123 (1979) 442; 125

(1980) 117) bound the $SU(2)$ Wilson loop by a $\mathbb{Z}_2$ gauge theory's through the decomposition $SU(2) \rightarrow \mathbb{Z}_2 \times SU(2)/\mathbb{Z}_2$, so confinement in $SU(2)$ follows wherever the $\mathbb{Z}_2$ theory confines; the four-dimensional $\mathbb{Z}_2$ theory deconfines at weak coupling, being dual to the Ising model, so the comparison reaches only strong coupling, and the same holds for $SU(3)$ against $\mathbb{Z}_3$. Tomboulis and Yaffe (Commun. Math. Phys. 100 (1985) 313) derive exact inequalities between electric-flux and vortex free energies from reflection positivity and the $\mathbb{Z}_N$ Fourier structure, and use them to prove deconfinement at high temperature. A claimed proof of confinement for all couplings in $SU(2)$ by these methods with approximate decimations (Tomboulis, arXiv:0707.2179) has not been established as a theorem in the literature as far as this author knows. All of these concern the string tension, that is, the energy of the electric-flux sectors, which on a torus is the 27-sector structure of the $SU(3)$ constants §4 and which disappears in infinite volume. The Jaffe–Witten gap is the vacuum-sector gap, the glueball mass, and no rigorous inequality relates it to the string tension in either direction. So the centre-based inequalities are real, non-abelian, and aimed at a different quantity, and their comparison theory has a transition of its own.

2. Reflection-positivity tools

Chessboard estimates (Fröhlich, Israel, Lieb and Simon, Commun. Math. Phys. 62 (1978) 1) hold for the Wilson measure and bound the probability of a local event by a free-energy difference per block, uniformly in the volume and at every coupling. They give rigorous large-deviation bounds for bad blocks without an expansion. They do not give mixing on the good blocks, so they are an ingredient of a verification and not a substitute for it.

Infrared bounds (Fröhlich, Simon and Spencer, Commun. Math. Phys. 50 (1976) 79) need Gaussian domination, $Z(h) \leq Z(0)$ for shifts h of the field, which uses the translation structure of a flat target space or of $U(1)$; on a non-abelian group manifold the shift is not a symmetry and the domination fails. This is consistent with confinement: an infrared bound would prove a massless phase.

Reflection positivity proves transitions (Peierls-type arguments via chessboard) and, through the transfer matrix, converts decay into a gap (Dobrushin note §3). It does not prove the absence of a transition.

3. Hamiltonian finite-size criteria

Knabe's criterion (J. Stat. Phys. 52 (1988) 627) and its descendants bound the gap of an infinite frustration-free Hamiltonian by the gap on finite blocks. The Kogut–Susskind Hamiltonian is not frustration-free: its ground state is not a ground state of the electric and magnetic terms separately, and no equivalent frustration-free form with the same gap is known. Ground-state clustering also does not imply a gap for a general Hamiltonian, while Euclidean decay in time does, through the positive transfer matrix; this is why the decay-to-gap step in this programme is always taken on the Euclidean side.

4. Exact transformations

Flow conjugation is exact and preserves the spectrum at the same coupling (flow-conjugation note). *Electric–magnetic duality* is exact for abelian groups and maps strong to weak coupling; for $SU(N)$ the dual is the signed surface gas of Section 1. *Gauge fixing* changes the specification, so the Dobrushin coefficient can be reduced by removing fixed links from the neighbour count, by at most the fraction of links on a maximal tree, about one quarter in four dimensions; this moves 444 to about 330 and touches nothing else.

5. Other constructions

Large N has $1/N$ as a small parameter; the goal is $SU(3)$. *Stochastic quantization* (the Langevin dynamics of the gauge field) constructs the two- and three-dimensional theories locally in time and says nothing about the gap; four dimensions is not accessible. *Small volume* relocates the crossover from the coupling to the volume variable: the small-volume theory has an explicit gap $\delta_1 g^{2/3} \hbar c / L$, and the passage to $L\Lambda \gg 1$ is the same confinement-scale problem.

6. The confinement scale as a single lattice theory

After the small-field renormalization has run to the scale where the effective coupling is of order one, the problem is one lattice gauge theory at that scale with a bounded but unknown effective interaction, to be shown mixing uniformly in the volume. Beyond that scale the block variables decorrelate and the effective interaction approaches the trivial fixed point, inside the completely analytical set. The whole difficulty is therefore concentrated at one scale and one effective interaction, which is the sharpest form of the statement that the problem has no small parameter, and which is exactly what the block verification of item 2 addresses.

7. Consequence for STATE

The review is closed. By written derivation, the confinement-scale region is entered by nothing known; the two unblocking routes are a correlation inequality for $SU(3)$, which would be new mathematics, and the computer-assisted block verification, which is a decision about the repository's rules and a large project. Constant-tightening on either side of the region remains available and does not touch it.